

Notes: Arnold, Dobbie, and Yang (QJE, 2018)
Racial Bias in Bail Decisions

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1 Big picture

- **Question.** Do racial disparities in bail decisions reflect accurate statistical discrimination, or do they reflect racial bias against black defendants?
- **Setting.** The paper studies bail hearings in Philadelphia and Miami-Dade. These are high-stakes, rapid decisions in which judges decide whether a defendant should be released before trial and under what conditions.
- **Core empirical problem.** Average rearrest rates by race do not reveal racial bias because black and white defendants can have different risk distributions. This is the inframarginality problem.
- **Theoretical idea.** The paper adapts Becker’s outcome test to bail. If judges are racially unbiased, marginally released white and black defendants should have the same pretrial misconduct rate. If judges are biased against black defendants, marginally released white defendants should be riskier than marginally released black defendants.
- **Research design.** The paper uses quasi-random assignment of defendants to bail judges. Judge leniency serves as an instrument for pretrial release, allowing the authors to estimate misconduct rates for defendants at the margin of release.
- **Main empirical result.** Marginally released white defendants are about 22 to 23 percentage points more likely to be rearrested before disposition than marginally released black defendants. This is evidence that black defendants are overdeterred at the margin.
- **Mechanism.** The evidence is more consistent with racially biased prediction errors or stereotypes than with pure racial animus. Bias is larger among part-time and inexperienced judges, and black defendants are overrepresented in the right tail of the predicted-risk distribution.
- **Economic takeaway.** The paper shows how to test bias at the decision margin rather than in averages. In this setting, observed racial disparities in bail cannot be explained by accurate statistical discrimination alone.

2 Incremental contribution of the paper

- **From average outcome tests to marginal outcome tests.** Earlier outcome tests compare average success or failure rates across groups. This paper argues that the correct comparison is at the margin of release, because bail judges choose a cutoff.
- **A new empirical test for absolute racial bias.** The paper provides a test that can detect absolute bias against black defendants, not merely relative bias across judge races.
- **Using judge quasi-random assignment to solve inframarginality.** The key innovation is to use judge leniency as an instrument to estimate race-specific local treatment effects for marginal defendants.
- **Two complementary estimators.** The paper develops both an IV estimator and a marginal treatment effects (MTE) estimator. IV requires fewer functional-form assumptions; MTE allows judge-specific margins and equal weighting across judges.

- **A high-stakes criminal justice application.** Bail is an unusually clean setting because the legal objective is relatively narrow: release most defendants while minimizing pretrial misconduct.
- **Mechanism evidence.** The paper goes beyond documenting bias by studying whether the bias is more consistent with racial animus or racially biased prediction errors. Evidence from judge race, judge experience, and predicted-risk distributions points toward inaccurate stereotypes.
- **Policy contribution.** The findings speak directly to bail reform, judicial training, feedback, and data-based risk assessment. The paper suggests that racial disparities may fall if judges make fewer prediction errors.

3 Data and measurement

3.1 Court data and jurisdictions

- The paper uses administrative court data from Philadelphia and Miami-Dade.
- Philadelphia records cover defendants arrested and charged between 2010 and 2014.
- Miami-Dade records cover defendants arrested and charged between 2006 and 2014.
- The final estimation sample contains 162,836 cases from Philadelphia and 93,417 cases from Miami-Dade, for 256,253 cases in total.
- The data contain defendant race, gender, age, date of birth, zip code, charges, offense severity, bail type, bail amount, bail judge, court appearances, and subsequent criminal justice outcomes.

Interpretation: The data are unusually well suited to this question because they connect the initial bail decision to both judge assignment and later pretrial misconduct.

3.2 Bail decisions

- Judges can release defendants on recognizance, assign nonmonetary conditions, set monetary bail, or deny bail for the most serious cases.
- The empirically central margin is release versus detention before trial.
- The paper treats pretrial release as the treatment whose effect on misconduct is estimated at the margin.
- Bail hearings are fast, often held by video, and conducted with limited information. This makes the setting especially relevant for prediction errors and heuristic decision-making.

Interpretation: The narrow legal objective of bail makes it easier to define an outcome test than in sentencing, where judges may simultaneously pursue punishment, deterrence, rehabilitation, incapacitation, and mercy.

3.3 Outcome variables

- The main outcome is rearrest before case disposition.
- The paper also studies rearrest by crime type: drug, property, and violent crimes.
- In Philadelphia, the data also include failures to appear in court.
- The paper checks robustness using broader misconduct measures that combine rearrest and failure to appear where possible.

Interpretation: The outcome test requires a measurable failure outcome. Rearrest is imperfect because arrests may depend on policing behavior, but it is the most consistently observed pretrial misconduct measure across the two jurisdictions.

3.4 Judge leniency instrument

- Judge leniency is measured using a residualized leave-out judge release rate.
- The measure removes court-by-time fixed effects so that comparisons are made within the quasi-random assignment cells.
- The instrument is constructed separately by defendant race and year. This allows judges to differ in how lenient they are toward white and black defendants.
- In the paper’s sample, moving from the least to the most lenient judge changes the release probability by about 53.6 percentage points after residualization.

Interpretation: The leave-out design prevents a defendant’s own release decision from mechanically affecting the leniency measure used to instrument that same defendant’s release.

3.5 Raw racial disparities

- Black defendants are more likely to face monetary bail and less likely to be released pretrial than white defendants.
- Conditional on release status, black and white defendants also differ in charge mix, prior records, and observed pretrial outcomes.
- Released black defendants have higher average rearrest rates than released white defendants, but this average comparison is not a valid test of judge bias.

Interpretation: The raw disparities motivate the paper, but they do not identify racial bias because average defendant risk and marginal defendant risk are different objects.

4 Models and empirical specifications

4.1 Becker-style outcome test

Let i index defendants, j index judges, and $r \in \{W, B\}$ denote defendant race. The expected cost of release is written as the expected probability of pretrial misconduct, $E[\alpha_i \mid r_i = r]$, with the cost of misconduct normalized to one. Judge j releases defendant i when

$$E[\alpha_i \mid r_i = r] \leq t_r^j(V_i),$$

where $t_r^j(V_i)$ is the judge's perceived benefit of release for defendants of race r with observable characteristics V_i .

- Judge j is biased against black defendants if $t_W^j(V_i) > t_B^j(V_i)$.
- The marginal defendant for judge j and race r is the defendant for whom the release cost equals the release threshold.
- If a judge is biased against black defendants, the marginal white defendant has a higher true misconduct risk than the marginal black defendant:

$$\alpha_W^j > \alpha_B^j.$$

Interpretation: The test is not whether average black and white defendants have equal misconduct. The test is whether black and white defendants at the judge's release margin have equal misconduct.

4.2 Racially biased prediction errors

The paper also develops an equivalent prediction-error interpretation. Judges may share the same objective but misperceive risk by race:

$$E^j[\alpha_i \mid V_i, r_i] = E[\alpha_i \mid V_i, r_i] + \tau_r^j(V_i).$$

- If $\tau_B^j(V_i) > \tau_W^j(V_i)$, the judge overpredicts black defendants' risk relative to white defendants' risk.
- This produces the same empirical prediction as taste-based bias: marginally released whites will be riskier than marginally released blacks.
- The paper later uses mechanism evidence to distinguish prediction errors from animus.

Interpretation: The same outcome-test statistic can reveal bias, but it cannot by itself say whether the bias comes from animus or from inaccurate racial beliefs.

4.3 IV estimator

The paper estimates the effect of pretrial release on misconduct separately for white and black defendants:

$$\begin{aligned} Y_{itj} &= \alpha_W^{IV} \text{Released}_{itj} + \beta_W X_{it} + v_{itj}, \\ Y_{itj} &= \alpha_B^{IV} \text{Released}_{itj} + \beta_B X_{it} + v_{itj}. \end{aligned}$$

- Y_{itj} is pretrial misconduct, usually rearrest before disposition.
- $Released_{itj}$ is instrumented using judge leniency.
- X_{it} includes court-by-time fixed effects and baseline defendant and charge controls.
- The IV estimate of racial bias is

$$D^{IV} = \alpha_W^{IV} - \alpha_B^{IV}.$$

Interpretation: The IV estimator recovers a weighted average of the race-specific treatment effects for defendants near the margin of release. The key comparison is the white LATE minus the black LATE.

4.4 MTE estimator

The MTE approach estimates race-specific marginal treatment effects over the support of the judge-induced propensity score:

$$MTE_W(p_W^j) = \frac{\partial}{\partial p_W^j} E(\ddot{Y}_{itj} \mid p_W^j, W),$$

$$MTE_B(p_B^j) = \frac{\partial}{\partial p_B^j} E(\ddot{Y}_{itj} \mid p_B^j, B).$$

The equal-weighted MTE estimate of racial bias is

$$D^{MTE} = \frac{1}{J} \sum_{j=1}^J [MTE_W(p_W^j) - MTE_B(p_B^j)].$$

- IV weights depend on the distribution of compliers induced by the instrument.
- MTE lets the paper average judge-specific margins using equal judge weights.
- The cost is that MTE requires more functional-form structure because judge leniency is discrete rather than continuously distributed.

Interpretation: The IV and MTE estimates answer closely related questions. Their similarity strengthens the paper’s main conclusion because it shows the finding is not driven by a peculiar weighting scheme.

4.5 Robustness and mechanism specifications

- The paper estimates results for different crime-type outcomes.
- It checks robustness to alternative misconduct definitions, including failure to appear where available.
- It reweights observable characteristics to compare black defendants to observationally similar white defendants.

- It compares black defendants to non-Hispanic white defendants.
- It controls for monetary bail amount to test whether money bail has an independent effect on misconduct.
- It estimates heterogeneity by judge specialization, judge experience, and judge race to study mechanisms.

Interpretation: These checks are important because the outcome test could be contaminated by omitted objectives, racial differences in arrest probability, or differences in judge behavior across institutional settings.

5 Identification and empirical design

5.1 What the design can identify

- The design identifies misconduct risk for defendants at the margin of release, not for the average released defendant.
- The IV estimate is a local average treatment effect for defendants whose release status changes with judge leniency.
- The MTE estimate reconstructs judge-specific release margins and compares them by race.
- The bias parameter is the difference between marginal white and marginal black misconduct rates.

Interpretation: The design is built around the decision rule of judges. A judge can be biased even if average released black defendants have higher rearrest rates than average released white defendants.

5.2 Quasi-random judge assignment

- Philadelphia uses a rotation system in which judges work shifts.
- Miami-Dade weekend bail hearings are also assigned through a rotation structure.
- The paper drops cases that do not follow the quasi-random assignment process.
- Court-by-time fixed effects absorb the assignment cells so that judge leniency is compared within the relevant rotation structure.

Interpretation: The credibility of the design depends on defendants not being sorted to lenient or strict judges after conditioning on court-by-time cells.

5.3 First-stage condition

- Judge leniency strongly predicts pretrial release.

- The first-stage coefficients are positive and statistically significant for all defendants, white defendants, and black defendants.
- The first stage is not one-for-one because the leniency measure is estimated with noise and because release decisions depend on case facts as well as judge tendencies.

Interpretation: Without a strong first stage, judge assignment would not provide meaningful variation in pretrial release. Table II and Figure I show that it does.

5.4 Exclusion restriction

- The exclusion restriction requires judge assignment to affect pretrial misconduct only through pretrial release.
- This is plausible because bail judges mainly make the bail decision and the outcome is measured before case disposition.
- The assumption would be threatened if judges affected later outcomes through other channels, such as attorney assignment, trial judge assignment, or bail conditions independent of release.

Interpretation: The exclusion restriction is not directly testable. The paper defends it using institutional details and robustness evidence from related work.

5.5 Monotonicity

- Monotonicity means that defendants released by stricter judges would also be released by more lenient judges.
- The paper allows judge leniency to vary by race, which relaxes the assumption that a judge's ranking is identical for white and black defendants.
- Within race, the paper presents evidence consistent with monotonicity across defendant and case subgroups.

Interpretation: Race-specific instruments matter because the paper is explicitly testing whether judges treat races differently at the margin.

5.6 Why OLS is not enough

- OLS compares average released and detained defendants, conditional on observables.
- Bail judges observe information that the econometrician may not observe.
- More importantly, the average treatment effect is not the marginal treatment effect unless risk distributions are identical or treatment effects are constant.
- This is the inframarginality problem: average outcomes can point in the wrong direction even when marginal outcomes reveal bias.

Interpretation: The paper’s central empirical contribution is to replace average comparisons with margin-specific comparisons.

6 Tables and figures

6.1 Table I: Descriptive Statistics

Read:

- Table I splits defendants by release status and race.
- Panel A describes bail type. Black defendants are less likely to receive release on recognizance and more likely to receive monetary bail.
- Panel B reports defendant characteristics such as gender, age, recent prior offenses, recent arrest while on bail, and recent failure to appear.
- Panel C reports charge characteristics. Black defendants have a higher share of felony and violent charges in the raw data.
- Panel D reports outcomes. Released black defendants have higher average rearrest and failure-to-appear rates than released white defendants.

Economic message: Table I motivates the analysis by showing large racial differences in bail conditions and outcomes. But these raw averages do not identify bias, because the outcome test must compare defendants at the release margin.

TABLE I
DESCRIPTIVE STATISTICS

	All defendants		White		Black	
	Released	Detained	Released	Detained	Released	Detained
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Bail type						
Release on recognizance	0.258	0.000	0.269	0.000	0.249	0.000
Nonmonetary bail w/conditions	0.195	0.030	0.203	0.033	0.189	0.028
Monetary bail	0.547	0.970	0.527	0.967	0.562	0.972
Bail amount (\$1,000s)	13.235	35.286	11.957	24.782	14.180	42.227
Panel B: Defendant characteristics						
Male	0.811	0.893	0.796	0.890	0.822	0.895
Age at bail decision	33.911	35.092	34.070	36.296	33.794	34.296
Prior offense in past year	0.287	0.466	0.272	0.464	0.299	0.466
Arrested on bail in past year	0.185	0.262	0.181	0.256	0.188	0.266
Failed to appear in court in past year	0.071	0.057	0.070	0.054	0.071	0.059
Panel C: Charge characteristics						
Number of offenses	2.722	3.162	2.544	2.587	2.854	3.541
Felony offense	0.482	0.538	0.450	0.473	0.506	0.581
Misdemeanor only	0.518	0.462	0.550	0.527	0.494	0.419
Any drug offense	0.390	0.260	0.373	0.244	0.403	0.271
Any DUI offense	0.084	0.007	0.091	0.007	0.079	0.007
Any violent offense	0.310	0.331	0.288	0.241	0.326	0.390
Any property offense	0.238	0.387	0.237	0.406	0.239	0.376
Panel D: Outcomes						
Rearrest prior to disposition	0.237	0.042	0.226	0.037	0.245	0.045
Rearrest drug crime	0.111	0.006	0.106	0.005	0.115	0.006
Rearrest property crime	0.086	0.022	0.082	0.022	0.089	0.022
Rearrest violent crime	0.078	0.021	0.061	0.013	0.091	0.026
Failure to appear in court (Phl only)	0.258	0.006	0.250	0.006	0.264	0.007
Failure to appear in court or rearrest	0.348	0.044	0.325	0.039	0.366	0.048
Observations	178,765	77,488	76,015	30,831	102,750	46,657

Notes. This table reports descriptive statistics for the sample of defendants from Philadelphia and Miami-Dade counties. The sample consists of bail hearings that were quasi-randomly assigned from Philadelphia between 2010 and 2014 and from Miami-Dade between 2006 and 2014, as described in the text. Information on race, gender, age, and criminal outcomes is derived from court records. Released is defined as being released at any point before trial. Detained is defined as never being released before trial. Bail amount (in \$1,000s) includes zeros. Failure to appear in court is defined only in Philadelphia. See [Online Appendix D](#) for additional details on the sample and variable construction.

Table I: Descriptive Statistics.

6.2 Figure I: First-Stage and Reduced-Form Results

Read:

- Panels A, C, and E show the first-stage relationship between judge leniency and residualized

pretrial release.

- The first stage is increasing for all defendants, white defendants, and black defendants.
- Panels B, D, and F show reduced-form relationships between judge leniency and pretrial misconduct.
- The reduced-form misconduct relationship is much stronger for white defendants than for black defendants.

Economic message: Figure I gives the visual version of the design. Lenient judges release more defendants of both races, but the additional released white defendants generate more pretrial rearrests than the additional released black defendants.

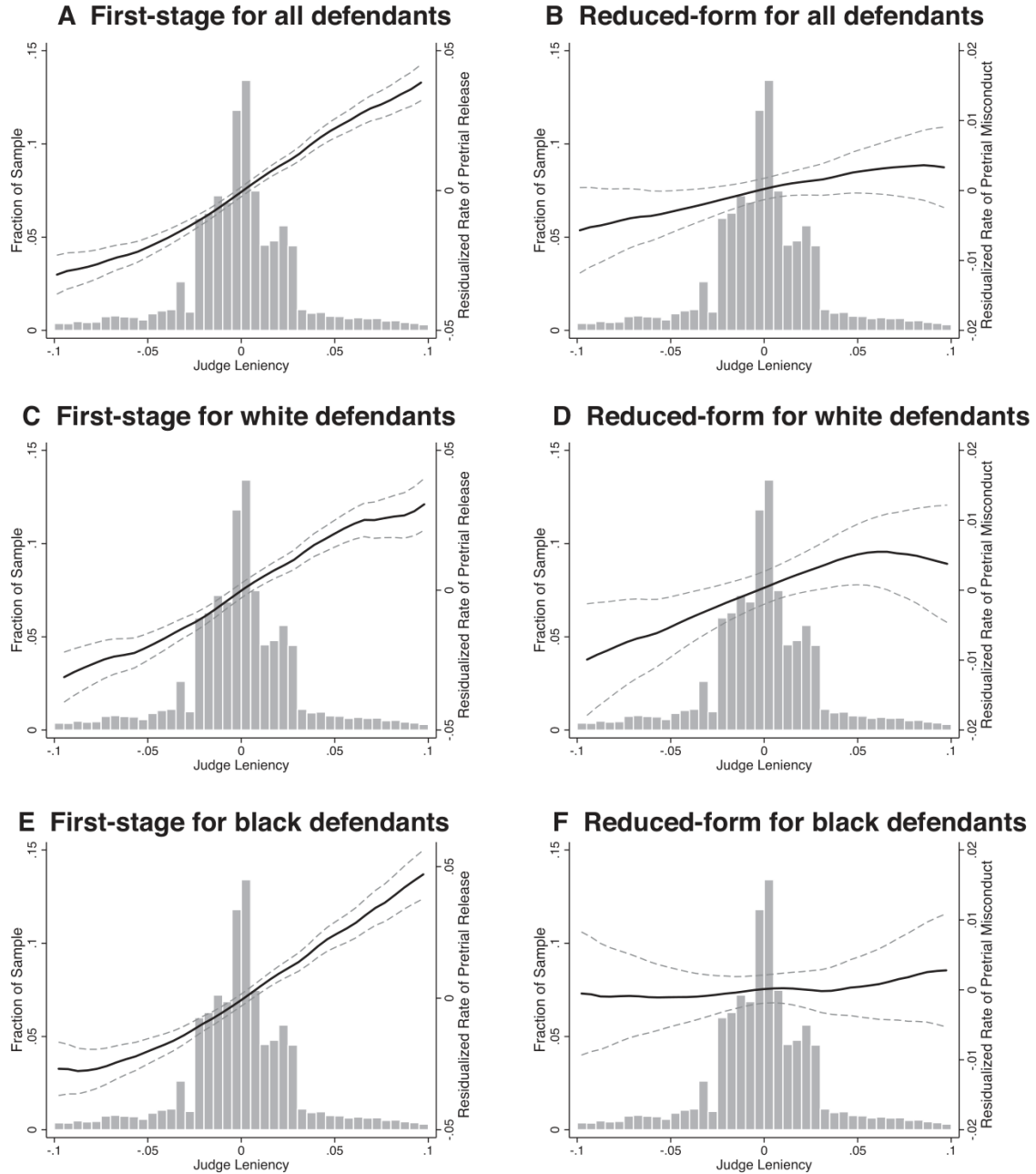


FIGURE I

First-Stage and Reduced-Form Results

Figure I: First-Stage and Reduced-Form Results.

6.3 Table II: First-Stage Results

Read:

- Table II reports the formal first-stage estimates.

- With baseline controls, a 10 percentage point increase in judge leniency raises pretrial release by about 3.89 percentage points for all defendants.
- The corresponding first-stage estimates are about 3.60 percentage points for white defendants and 4.15 percentage points for black defendants.
- All estimates are statistically significant at the 1% level.

Economic message: The table establishes that judge leniency is a powerful instrument for pretrial release. It also shows that the instrument works separately within race.

TABLE II
FIRST-STAGE RESULTS

	All defendants		White		Black	
	(1)	(2)	(3)	(4)	(5)	(6)
Pretrial Release	0.405*** (0.027) [0.698]	0.389*** (0.025) [0.698]	0.373*** (0.036) [0.711]	0.360*** (0.032) [0.711]	0.434*** (0.036) [0.688]	0.415*** (0.033) [0.688]
Court x year FE	Yes	Yes	Yes	Yes	Yes	Yes
Baseline controls	No	Yes	No	Yes	No	Yes
Observations	256,253	256,253	106,846	106,846	149,407	149,407

Notes. This table reports the first-stage relationship between pretrial release and judge leniency. The regressions are estimated on the sample as described in the notes to Table I. Judge leniency is estimated using data from other cases assigned to a bail judge in the same year, constructed separately by defendant race, following the procedure described in Section III.B. All regressions include court-by-time fixed effects. Baseline controls include race, gender, age, whether the defendant had a prior offense in the past year, whether the defendant had a prior history of pretrial crime in the past year, whether the defendant had a prior history of failure to appear in the past year, the number of charged offenses, indicators for crime type (drug, DUI, property, violent, and other), crime severity (felony and misdemeanor), and indicators for any missing controls. The sample mean of the dependent variable is reported in brackets. Robust standard errors two-way clustered at the individual and judge-by-shift level are reported in parentheses. ***Significant at 1% level.

Table II: First-Stage Results.

6.4 Table III: Test of Randomization

Read:

- Table III compares whether defendant and charge characteristics predict actual release and whether they predict assigned judge leniency.
- Defendant and charge characteristics strongly predict actual release, as expected.

- The same characteristics do not jointly predict judge leniency after the court-by-time controls, which supports quasi-random judge assignment.
- The table is rotated in the original paper; the extracted image below is rotated for readability.

Economic message: The table addresses the core selection concern. Defendants look very different in their release probabilities, but they do not appear systematically sorted to more lenient or stricter judges within assignment cells.

TABLE III
TEST OF RANDOMIZATION

	All		White		Black	
	Pretrial release (1)	Judge leniency (2)	Pretrial release (3)	Judge leniency (4)	Pretrial release (5)	Judge leniency (6)
Male	-0.09424*** (0.00235)	-0.00005 (0.00024)	-0.08593*** (0.00325)	0.00004 (0.00038)	-0.10379*** (0.00323)	-0.00014 (0.00031)
Age at bail decision	-0.01725*** (0.00086)	-0.00009 (0.00009)	-0.02250*** (0.00127)	-0.00015 (0.00016)	-0.01512*** (0.00104)	-0.00005 (0.00010)
Prior offense in past year	-0.14922*** (0.00287)	-0.00017 (0.00028)	-0.16817*** (0.00445)	0.00030 (0.00046)	-0.13411*** (0.00362)	-0.00044 (0.00036)
Arrested on bail in past year	0.01066*** (0.00355)	0.00004 (0.00034)	0.01967*** (0.00552)	-0.00166*** (0.00057)	0.00495 (0.00439)	0.00116*** (0.00042)
Failed to appear in court in past year	0.03318*** (0.00413)	0.00012 (0.00025)	0.03253*** (0.00631)	0.00104** (0.00043)	0.03245*** (0.00529)	-0.00047 (0.00031)
Number of offenses	-0.02090*** (0.00053)	-0.00001 (0.00003)	-0.01829*** (0.00085)	-0.00002 (0.00006)	-0.02131*** (0.00063)	0.00000 (0.00004)
Felony offense	-0.17618*** (0.00257)	-0.00003 (0.00012)	-0.18817*** (0.00397)	-0.00014 (0.00020)	-0.16948*** (0.00323)	0.00004 (0.00014)
Any drug offense	0.03514*** (0.00258)	-0.00038 (0.00026)	0.02558*** (0.00357)	-0.00002 (0.00039)	0.04069*** (0.00332)	-0.00063* (0.00032)
Any property offense	-0.04272*** (0.00285)	-0.00013 (0.00026)	-0.05560*** (0.00388)	0.00009 (0.00041)	-0.03188*** (0.00354)	-0.00029 (0.00033)
Any violent offense	0.01640*** (0.00389)	0.00028 (0.00025)	0.07515*** (0.00497)	0.00033 (0.00045)	-0.02443*** (0.00429)	0.00023 (0.00029)
Joint <i>F</i> -test	[.00000]	[.60067]	[.00000]	[.21951]	[.00000]	[.08289]
Observations	256,253	256,253	106,846	106,846	149,407	149,407

Notes. This table reports reduced-form results testing the random assignment of cases to bail judges. The regressions are estimated on the sample as described in the notes to Table I. Judge leniency is estimated using data from other cases assigned to a bail judge in the same year, constructed separately by defendant race, following the procedure described in Section III.B. Columns (1), (3), and (5) report estimates from an OLS regression of pretrial release on the variables listed and court-by-time fixed effects. Columns (2), (4), and (6) report estimates from an OLS regression of judge leniency on the variables listed and court-by-time fixed effects. The *p*-value reported at the bottom of the columns is for an *F*-test of the joint significance of the variables listed in the rows. Robust standard errors two-way clustered at the individual and the judge-by-shift level are reported in parentheses. ***Significant at 1% level, **significant at 5% level, *significant at 10% level.

Table III: Test of Randomization.

6.5 Table IV: Pretrial Release and Criminal Outcomes

Read:

- Table IV contains the paper’s core racial-bias estimates.
- In the IV results, pretrial release raises rearrest for marginal white defendants by 23.6 percentage points, but by only 1.4 percentage points for marginal black defendants.
- The IV estimate of racial bias is therefore 22.2 percentage points.
- In the MTE results, the corresponding bias estimate is 23.1 percentage points.

- The estimates by crime type suggest especially strong differences for property crimes, with positive but less precise differences for drug and violent crimes.

Economic message: Marginally released white defendants are much riskier than marginally released black defendants. Under the model, that is evidence that judges set a tougher release threshold for black defendants.

TABLE IV
PRETRIAL RELEASE AND CRIMINAL OUTCOMES

	IV results			MTE results		
	White (1)	Black (2)	D^{IV} (3)	White (4)	Black (5)	D^{MTE} (6)
Panel A: Rearrest for all crimes						
Rearrest prior to disposition	0.236*** (0.073) [0.172]	0.014 (0.070) [0.182]	0.222** (0.101) —	0.249*** (0.084) [0.172]	0.017 (0.080) [0.182]	0.231** (0.117) —
Panel B: Rearrest by crime type						
Rearrest for drug crime	0.067 (0.043) [0.077]	0.019 (0.043) [0.081]	0.047 (0.060) —	0.074 (0.048) [0.077]	−0.024 (0.054) [0.081]	0.097 (0.074) —
Rearrest for property crime	0.158*** (0.057) [0.065]	−0.005 (0.047) [0.068]	0.163** (0.073) —	0.149** (0.066) [0.065]	0.043 (0.053) [0.068]	0.106 (0.084) —
Rearrest for violent crime	0.079** (0.039) [0.047]	−0.000 (0.042) [0.071]	0.080 (0.058) —	0.082* (0.044) [0.047]	−0.001 (0.050) [0.071]	0.083 (0.068) —
Observations	106,846	149,407	—	106,846	149,407	—

Notes. This table reports estimates of racial bias in pretrial release based on rearrest prior to case disposition. Columns (1) and (2) report two-stage least squares results of the impact of pretrial release on the probability of pretrial misconduct separately by race, and column (3) reports the difference between the white and black two-stage least squares coefficients, or D^{IV} as described in the text. Columns (1)–(3) use instrumental variables weights for each specification and report robust standard errors two-way clustered at the individual and judge-by-shift level in parentheses. Columns (4) and (5) report the average marginal treatment effect of the impact of pretrial release on the probability of pretrial misconduct separately by race, and column (6) reports the difference between the white and black MTE coefficients, or D^{MTE} as described in the text. Columns (4)–(6) use equal weights for each judge and report bootstrapped standard errors clustered at the judge-by-shift level in parentheses. All specifications control for court-by-time fixed effects and defendant race, gender, age, whether the defendant had a prior offense in the past year, whether the defendant had a prior history of pretrial crime in the past year, whether the defendant had a prior history of failure to appear in the past year, the number of charged offenses, indicators for crime type (drug, DUI, property, violent, or other), crime severity (felony or misdemeanor), and indicators for any missing characteristics. The sample means of the dependent variables are reported in brackets. ***Significant at 1% level, **significant at 5% level, *significant at 10% level.

Table IV: Pretrial Release and Criminal Outcomes.

6.6 Figure II and Figure III: Marginal Treatment Effects and Predicted Risk

Read:

- Figure II plots the race-specific MTE curves. The white curve lies above the black curve over the judge-leniency support.
- Figure III plots the predicted risk distribution by race and the representativeness ratio for black versus white defendants.
- Black defendants are overrepresented in the right tail of the predicted-risk distribution.
- The sharp right-tail pattern is consistent with a stereotyping model in which judges overweight the most distinctive differences between groups.

Economic message: Figure II shows that the bias result is not only an average-weighting artifact. Figure III helps explain the mechanism: if judges focus on the right tail of risk, they may exaggerate the danger of releasing black defendants.

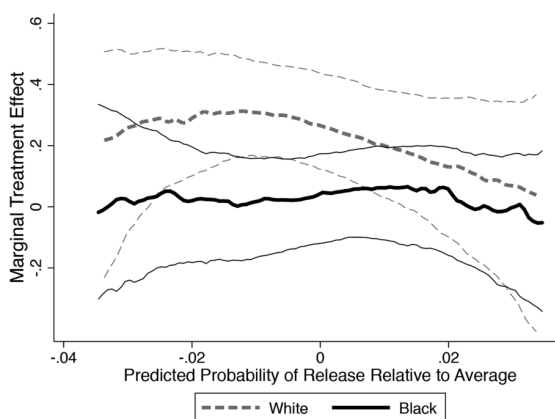


FIGURE II
Marginal Treatment Effects

(a) Figure II: Marginal Treatment Effects.

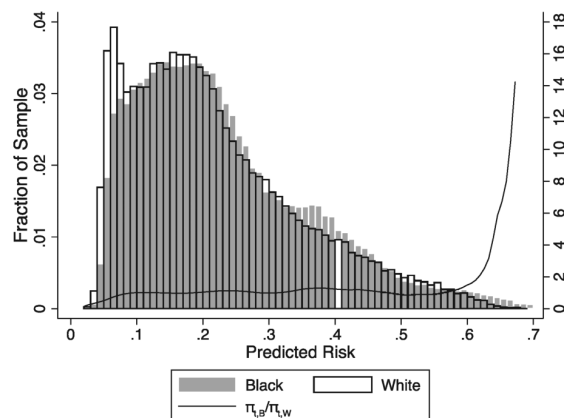


FIGURE III
Predicted Risk Distribution by Defendant Race

(b) Figure III: Predicted Risk Distribution by Defendant Race.

6.7 Table V: Racial Bias by Judge Experience

Read:

- Table V estimates racial bias separately by judge specialization and experience.
- Bias is larger in Miami-Dade, where weekend bail judges are part-time nonspecialists.
- Bias is much smaller and statistically insignificant in Philadelphia, where bail judges are full-time specialists.
- Among Miami-Dade judges, low-experience judges show very large bias estimates: 48.7 percentage points using IV and 51.0 percentage points using MTE.

- High-experience Miami-Dade judges have much smaller and statistically insignificant estimates.

Economic message: The heterogeneity pattern points toward prediction errors rather than pure animus. Experience and specialization appear to reduce the degree to which judges overdetain black defendants at the margin.

TABLE V
RACIAL BIAS IN PRETRIAL RELEASE BY JUDGE EXPERIENCE

	Judge specialization				Judge experience			
	Miami D^{IV} (1)	Miami D^{MTE} (2)	Phl D^{IV} (3)	Phl D^{MTE} (4)	Miami low Exp D^{IV} (5)	Miami low Exp D^{MTE} (6)	Miami high Exp D^{IV} (7)	Miami high Exp D^{MTE} (8)
Panel A: Rearrest for all crimes								
Rearrest prior to disposition	0.251** (0.114) [0.149]	0.249** (0.121) [0.149]	0.040 (0.184) [0.194]	0.078 (0.195) [0.194]	0.487** (0.237) [0.148]	0.510** (0.233) [0.148]	0.144 (0.178) [0.152]	0.036 (0.164) [0.152]
Panel B: Rearrest by crime type								
Rearrest for drug crime	0.053 (0.066) [0.057]	0.103 (0.077) [0.057]	0.008 (0.138) [0.092]	0.015 (0.150) [0.092]	0.141 (0.119) [0.057]	0.185 (0.138) [0.057]	-0.013 (0.101) [0.057]	0.006 (0.110) [0.057]
Rearrest for property crime	0.196** (0.084) [0.078]	0.127 (0.096) [0.078]	-0.031 (0.110) [0.060]	-0.014 (0.199) [0.060]	0.296** (0.140) [0.078]	0.293* (0.163) [0.078]	0.146 (0.111) [0.079]	0.035 (0.124) [0.079]
Rearrest for violent crime	0.082 (0.065) [0.050]	0.079 (0.075) [0.050]	0.065 (0.115) [0.067]	0.066 (0.119) [0.067]	0.204 (0.134) [0.048]	0.218* (0.119) [0.048]	0.032 (0.100) [0.051]	-0.036 (0.099) [0.051]
Observations	93,417	93,417	162,836	162,836	47,692	47,692	45,725	45,725

Notes. This table reports estimates of racial bias for different subgroups of judges. D^{IV} is the difference between the white and black two-stage least squares coefficient estimates of pretrial release on pretrial misconduct using instrumental variables weights. D^{MTE} is the difference between the average white and black MTE estimates of pretrial release on pretrial misconduct using equal weights by judge. Columns (1) and (2) report estimates for nonspecialist bail judges in Miami-Dade. Columns (3) and (4) report estimates for specialist bail judges in Philadelphia. Columns (5)–(8) report estimates for nonspecialist bail judges in Miami with below- and above-median years of experience. The sample is described in the notes to Table 1. The dependent variable is listed in each row. All specifications control for court-by-time fixed effects and defendant race, gender, age, whether the defendant had a prior offense in the past year, whether the defendant had a prior history of pretrial crime in the past year, whether the defendant had a prior history of failure to appear in the past year, the number of charged offenses, indicators for crime type (drug, DUI, property, violent, or other), crime severity (felony or misdemeanor), and indicators for any missing characteristics. The sample means of the dependent variables are reported in brackets. For IV specifications, robust standard errors two-way clustered at the individual and judge-by-shift level are reported in parentheses. For MTE specifications, bootstrapped standard errors clustered at the judge-by-shift level are reported in parentheses. **significant at 5% level, *significant at 10% level.

Table V: Racial Bias in Pretrial Release by Judge Experience.

7 Short summary

- The paper studies racial bias in bail decisions using administrative court data from Philadelphia and Miami-Dade.
- The key theoretical prediction is that unbiased judges should have equal misconduct rates for marginally released white and black defendants.
- Average comparisons are invalid because of inframarginality.
- The empirical strategy uses quasi-random judge assignment and judge leniency as an instrument for pretrial release.
- The IV estimate indicates that marginally released white defendants are 22.2 percentage points more likely to be rearrested than marginally released black defendants.

- The MTE estimate is very similar at 23.1 percentage points.
- Robustness checks show that the result is not driven by observable crime or defendant characteristics.
- Mechanism evidence points toward racially biased prediction errors or stereotypes: both white and black judges exhibit bias, black defendants are overrepresented in the right tail of predicted risk, and bias is greater among inexperienced and part-time judges.
- The paper’s main contribution is a method for testing bias at the margin of decision-making.

8 Conclusion

8.1 Core conclusion

Arnold, Dobbie, and Yang show that bail judges are racially biased against black defendants at the margin of release. The evidence is not that black defendants have lower average misconduct risk than white defendants. Instead, the evidence is that the marginal black defendants whom judges detain are safer than the marginal white defendants whom judges release. That is exactly the object required by the Becker-style outcome test.

8.2 Economic intuition

A judge’s bail decision can be viewed as a cutoff rule. Defendants below the judge’s perceived risk threshold are released; defendants above it are detained or face stricter bail conditions. If judges use the same risk threshold for black and white defendants, then the marginal released defendants of both races should have the same pretrial misconduct risk. The paper finds the opposite: marginal white defendants have much higher rearrest risk than marginal black defendants. This means that judges act as if black defendants need to be safer than white defendants to receive release.

8.3 Incremental contribution revisited

The paper advances the literature in three main ways. First, it turns the outcome test into a margin-specific empirical design that avoids the inframarginality problem. Second, it uses quasi-random judge assignment to estimate the relevant marginal misconduct rates. Third, it provides mechanism evidence showing that bias is plausibly linked to inaccurate stereotypes and prediction errors rather than only to explicit animus. This combination of theory, identification, and mechanism evidence makes the paper a major contribution to empirical work on discrimination in criminal justice.

8.4 Implications for bail policy and judicial decision-making

The findings imply that racial disparities in pretrial detention are not only a mechanical consequence of defendant risk. They also reflect judge decision-making errors at the margin. If those errors come from inaccurate stereotypes, then training, feedback, experience, and better risk information may reduce racial disparities. The results are also consistent with the broader argument that structured

prediction tools can sometimes improve human decisions, although such tools must themselves be evaluated for fairness, accuracy, and implementation risks.

8.5 Limitations

The analysis relies on rearrest and failure-to-appear outcomes, which are imperfect measures of true misconduct. If police are more likely to arrest black defendants conditional on actual misconduct, the paper’s estimates may understate bias against black defendants. The exclusion restriction cannot be tested directly. The MTE estimator requires additional functional-form assumptions. Finally, the evidence comes from two jurisdictions and may not generalize mechanically to all bail systems, especially where assignment processes or judge responsibilities differ.

8.6 Takeaway

The enduring takeaway is that tests of discrimination must look at the margin of decision-making. In bail, judges appear to set a stricter effective threshold for black defendants than for white defendants. The paper shows that racial bias can be uncovered by combining a simple economic model of decision thresholds with quasi-random variation in decision-maker leniency.